

Handling Multiple Variable

1. Background

The Horse Colic dataset (UCI Machine Learning Repository) contains medical records of horses experiencing colic (acute abdominal pain). This dataset consists of 300 rows and 28 columns, covering demographic data, vital signs, physical examination results, and laboratory test results. One of the main characteristics of this dataset is the high proportion of missing values in many clinical variables, which realistically represents the state of medical records in the field (not all tests are performed on every patient).

This case demonstrates the handling of missing data for two overlapping numeric variables, namely packed cell volume (PCV) and total protein (TP), using an Iterative Regression Imputation approach. The results are compared with a Complete Case Analysis (CCA) approach to determine its impact on regression model performance.

2. Methodology

2.1 Missing Data Identification

Initial examination revealed missing values for almost all clinical variables, ranging from less than 1% (surgery, outcome) to more than 80% (nasogastric reflux pH). The following table summarizes the variables with the highest missing values:

Variable	Number of Missing	% Missing
nasogastric_reflux_ph	247	83.3%
abdomo_protein	198	66.0%
abdomo_appearance	165	55.0%
abdomen	118	39.3%
nasogastric_reflux	106	35.3%
nasogastric_tube	104	34.7%
rectal_exam_feces	102	34.0%
total_protein	33	11.0%
packed_cell_volume	29	9.7%
pulse	24	8.0%

The analysis focused on packed_cell_volume and total_protein because they are both numerical, often analyzed together to assess dehydration/shock levels, and have 26 overlapping missing lines — conditions suitable for demonstrating this with iterative regression imputation.

2.2 Iterative Regression Imputation

Steps taken

1. **Starting value:** Simple random imputation — missing values in PCV and TP are initially imputed with a random sample of observed values for each variable.
2. Iterations: A total of 10 iterations, alternately:
 - a. Build a regression model *“packed_cell_volume ~ age + surgery + surgical_lesion + cp_data + total_protein_imp”* then predict & update the missing PCV values.
 - b. Build a regression model *“total_protein ~ age + surgery + surgical_lesion + cp_data + packed_cell_volume_imp”* (using the newly updated PCV), then predict and update the missing TP value.
3. The predictors used (age, surgery, surgical_lesion, cp_data) were chosen because they were fully observed (no missing values).
4. For comparison, a Complete Case Analysis (CCA) model was also built using only rows with complete PCV and TP (263 of 299 rows).

3. Result

3.1 Distribution of Imputation Values

The following comparison shows the distribution of imputation values for 26 simultaneously missing rows, between the initial conditions (simple random imputation) and after 10 iterations of iterative regression imputation.

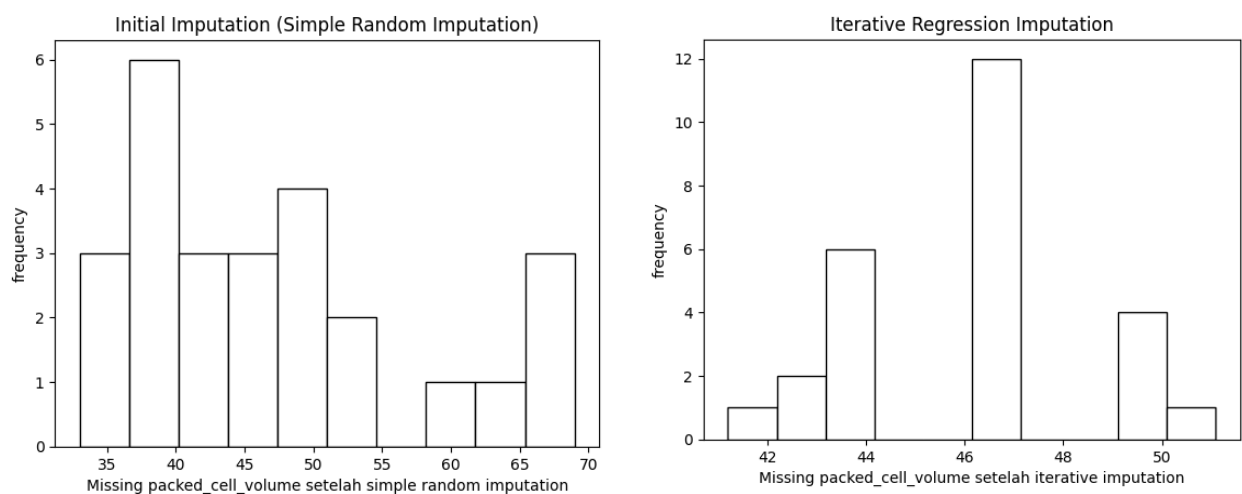


Figure 1. Distribution of packed_cell_volume imputation values — left: simple random imputation (initial), right: after iterative regression imputation.

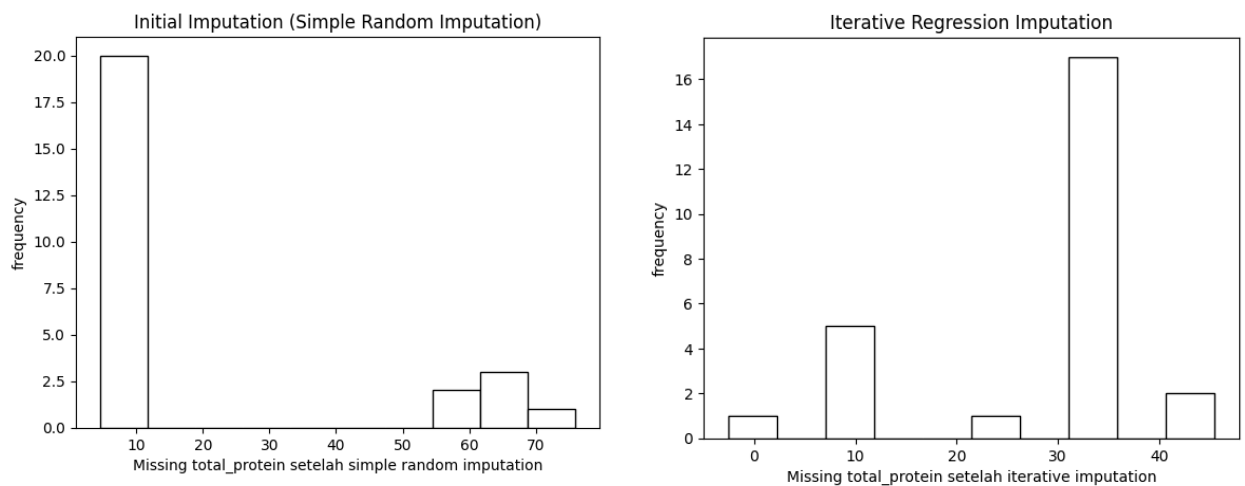


Figure 2. Distribution of total protein imputation values – left: simple random imputation (initial), right: after iterative regression imputation.

After iterative imputation, the imputed values are no longer purely random samples, but rather follow relationships (patterns) with other predictor variables. Consequently, their distribution changes and better reflects the actual data structure than random imputation alone. It should be noted that some total_protein values in the original data appear very high (above 60), likely representing outliers/anomalies in the source dataset, which may have influenced the model results.

3.2 Model Performance Comparison: CCA vs Iterative Imputation

The regression model “*packed_cell_volume ~ age + surgery + surgical_lesion + cp_data + total_protein*” was built on two dataset conditions: (1) Complete Case Analysis (263 rows) and (2) full dataset (299 rows) after iterative regression imputation.

Approach	Number of Rows	R-squared
Complete Case Analysis (CCA)	263	0.0626
Iterative Regression Imputation	299	0.0750

The model with iterative regression imputation yielded a slightly higher R-squared (0.0750 vs. 0.0626), while retaining a larger sample size for analysis (299 vs. 263 rows).

Standard error per coefficient:

Coefficient	CCA	Iterative
Intercept	3.1093	2.8108
age	2.2840	0.2602
surgery	1.6400	1.4604
surgical_lesion	1.6701	1.4898
cp_data	1.4797	1.3497
total_protein	0.0623	0.0249

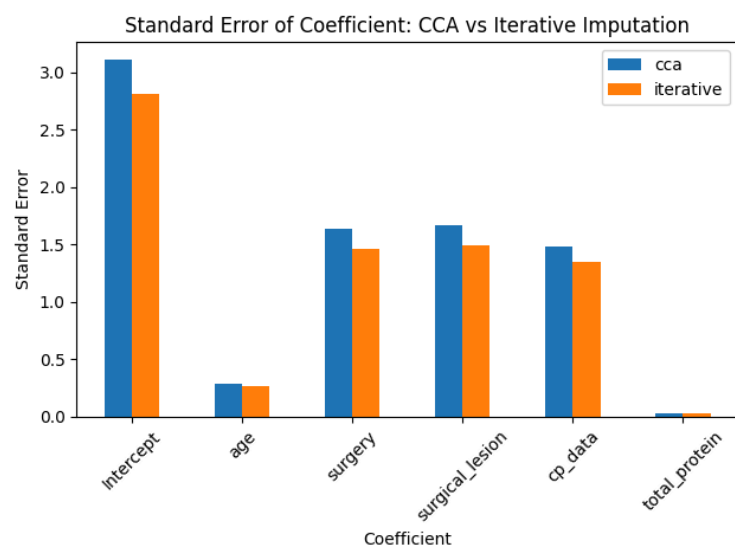


Figure 3. Comparison of standard error coefficients between CCA and Iterative Regression Imputation.

For all coefficients, the model with iterative regression imputation produces smaller standard errors than Complete Case Analysis, indicating more precise estimates—as the number of observations retained in the analysis increases.

4. Conclusion

- The Horse Colic dataset has widespread missing values across almost all clinical variables, with packed_cell_volume and total_protein being two examples of variables missing overlapping across 26 rows.
- Iterative Regression Imputation successfully filled in missing values for both variables simultaneously, utilizing the relationships between the variables and the complete set of predictors.
- Compared to Complete Case Analysis, the iterative imputation approach provided a slightly higher R-squared (0.0750 vs. 0.0626) and smaller standard errors across all coefficients, while maintaining a larger sample size (299 vs. 263 rows).
- It should be noted that the R-squared values for both approaches were still relatively low, indicating that the predictors used (age, surgery, surgical_lesion, cp_data) were not strong enough to explain clinical variation in packed_cell_volume. Adding other physiological predictors (if available and complete) has the potential to further improve model performance.